The fdamador neutrino: a report

Nov 4, 2026 · @FDA

Summary

Turning the toroidal photon's field inside out gives a neutral, self-sustaining ring of electric field, which matches two neutrino properties but cannot stay coherent long enough to be one. Hypothesis H28 is recorded as *bounded*.

- **What matches:** the inverted ring has zero charge and zero time-averaged magnetic moment, and its crossed circuit form keeps the half-integer (spin ½) modes.
- **What fails:** in vacuum it radiates away in about 3×10^{-16} s. SN1987A neutrinos survived about 5×10^{4} s of their own time, a gap of about 20 orders of magnitude.
- **The inflow:** the zero-point field of its own radiation resistance balances the outflow. The ring holds $\frac{1}{2}\hbar\omega_0$ forever with the energy ledger closed, but as random noise, not as a particle.
- **Missing entirely:** mass and the weak interaction, which is how every neutrino has been detected.

Every number below was computed in kronspice with an analytic twin beside it and a closed energy ledger.

The idea

The fdamador neutrino is the toroidal photon electron (Williamson and van der Mark, 1997) with its field inverted: the electric field forms the ring and the magnetic field circles round it. It was proposed by Franklin Amador on 4 November 2026 and entered in the kronspice hypothesis register as H28.

In the photon electron and the charged rings before it, a current runs round the ring and B wraps the tube. Here E runs round the ring and B wraps it. The proposal is that this inverted structure is the neutrino, so it is tested against the neutrino's measured properties rather than the electron's.

What Maxwell's equations say before any model

Three results follow from the field equations alone:

© 2026 fdamador@comcast.net

1. **No charge.** A closed loop of E has ∇·E = 0, so by Gauss's law it encloses no charge. It is neutral, as a neutrino is, and so it can never be an electron.
2. **No static form.** E can circulate round the ring only while the magnetic flux through the hole changes (Faraday's law). The ring sustains itself as an oscillation: E round the ring is a ring of displacement current, its B threads the hole, and the changing flux drives E.
3. **An oscillating magnetic moment.** The displacement-current ring is a magnetic dipole whose time average is zero. A time-varying magnetic dipole radiates, so the ring loses energy.

This field pattern already exists in engineering: it is the TE01δ mode of a dielectric ring resonator, used in microwave filters, where the dielectric holds the field in place.

Part 1: the E-ring resonator

A 10 mm ceramic ring with εr = 100 resonates at 2.0606 GHz with Q = 40.5, and kronspice agrees with the formula to four figures by two independent methods.

The ring is a lumped loop of 24 sections: series capacitors for the displacement path along the ring, the ring's self-inductance, a radiation resistance for its magnetic-dipole radiation, and dielectric loss. Radiation alone gives a closed-form Q:

```
Q_{rad} = \frac{6\Lambda}{\pi (ka)^3}, \qquad ka = \frac{\sqrt{2}\,(a/r)}
{\sqrt{\varepsilon_r \Lambda}}, \qquad \Lambda = \ln\frac{8a}{r} - \frac{7}
{4}
```

So Q grows as εr^1.5: **the permittivity is the confinement.** Bench ring: a = 10 mm, r = 2.5 mm, tan δ = 1×10⁻⁴.

εr	f_0 (GHz)	ka	Q, formula	Q, kronspice AC
30	3.762	0.788	6.68	6.68
100	2.061	0.432	40.52	40.52
300	1.190	0.249	207	207
1000	0.652	0.137	1,140	1,140

© 2026 fdamador@comcast.net

At εr = 100 a ring-down gives Q = 40.53. Its ledger closes: the 3.151×10^{-13} J kick equals 3.144×10^{-13} J radiated and heated plus 6.7×10^{-16} J still stored, with a time-stepping error of 2.5×10^{-18} J. No charge leaves the ring, and the moment averages to −0.007 of its peak. At εr = 30 the ring is no longer small against the wavelength (ka ≈ 0.8), so that row sits at the edge of the lumped model.

Part 2: the neutrino test bench

To be a neutrino the ring would need an effective vacuum permittivity of 3.4×10^{13}, and nothing known supplies one.

In vacuum (εr = 1, a/r = 4) the ring is electrically large at its own resonance (ka = 4.3), so the lumped model fails and gives Q ≈ 0.04: the field is gone within one period. The field solve of the uninverted photon loop shows the same, with light leaving in under half a period.

The bench asks what holding it would take. SN1987A neutrinos (about 10 MeV) crossed 168,000 light-years. At a declared, illustrative mass of 0.1 eV the Lorentz factor is 10^8, so the proper time is 5.3×10^4 s and the ring must keep its energy for $Q = 8\times10^{18}$.

Property	E-ring model	Measured or standard
Charge	0 (∇·E = 0)	consistent with 0
Spin ½	crossed inverted ring keeps half-integer modes (Part 3)	½
Static magnetic moment	0 (it oscillates)	< 2.8×10^{-11} μB (Borexino 2017); standard ≈ 3×10^{-20} μB at 0.1 eV
Lifetime in vacuum	≈ 3×10^{-16} s	⩾ 5.3×10^4 s proper time (SN1987A)
Confinement to survive	effective εr = 3.4×10^{13} at a = 1.5×10^{-12} m	none known
Mass	not given (0.1 eV declared)	< 0.45 eV (KATRIN 2025)
Weak interaction	not in the model	how every neutrino is detected

© 2026 fdamador@comcast.net

Part 3: the inverted travelling wave

Crossing the conductors of the inverted ring line still gives half-integer modes, so the spin-½ property of the photon electron survives the inversion.

Swapping E and B in a ring transmission line gives its circuit dual: series capacitors carry E along the path and shunt inductors carry B round it. That is a backward-wave ladder, where the longest wavelength has the highest frequency. Its modes follow $\omega = 1/(2\sqrt{(C \cdot Lg)} \cdot \sin(\pi k/N))$, with k whole on the plain ring and half-integer on the crossed one. Built in kronspice with 100 sections, C = 1 nF and Lg = 1 µH:

Mode	Plain ring (MHz)	Wavelengths	Crossed ring (MHz)	Wavelengths
1	80.11	1	160.21	½
2	40.08	2	53.42	1½
3	26.74	3	32.07	2½
4	20.08	4	22.93	3½

The AC peaks match the ladder formula to within 5×10^{-5}, and both AC ledgers balance.

Part 4: an inflow that balances the outflow

The zero-point field balances the ring's radiation exactly, so the ring never decays to zero, but what it holds is half a quantum of random noise.

By the fluctuation–dissipation theorem (Callen and Welton, 1951), the resistance that radiates the ring's energy is also a noise source: the field it radiates into drives it back. At the neutrino ring's 0.1 eV, thermal noise is negligible even in the 2.7 K cosmic background (occupation e^{-430}), leaving the zero-point part. Inflow and outflow are both booked on the magnetic-field channel, so they share one ledger column. Results from a scaled loop at the bench ring's radiation Q of 40.7, over 4 realizations:

Quantity	kronspice	Expected
Stored energy	$0.4941 \pm 0.0067\ \hbar\omega_0$	$0.4916\ \hbar\omega_0$ (½$\hbar\omega_0$ for the modes present)
Magnetic channel	391.77 J in = 389.36 J out + 2.15 J stored	in = out + stored

© 2026 fdamador@comcast.net

Quantity	kronspice	Expected
Energy fluctuation σ/⟨E⟩	0.97	≈ 1 for a random field
Mean magnetic moment	1.9×10^{-4} of its rms	0

Does the Casimir effect add inflow? No, but the boundaries that cause it change the outflow. The Casimir force comes from walls changing which zero-point modes fit between them, a fixed energy shift rather than a continuous supply. The same walls change the ring's radiation rate by a Purcell factor F, and the noise changes with it, so the held energy stays put:

Purcell factor F	Q	Stored, kronspice ($\hbar\omega_0$)	Expected ($\hbar\omega_0$)	Lifetime
4 (enhanced)	10.2	0.4794 ± 0.0023	0.4817	×¼
1 (free space)	40.7	0.4941 ± 0.0067	0.4916	×1
0.2 (suppressed)	203	0.4843 ± 0.0318	0.4942	×5

A coherent ring lasting SN1987A's proper time would need $F \approx 5\times10^{-21}$. The classic measurement of inhibited emission found about 20× (Hulet, Hilfer and Kleppner, 1985), and a free neutrino has no walls.

What the expected value caught. The first version read 8 % high at low Q. The noise started at a non-zero voltage, so the start-up operating point charged the series capacitors through 1 GΩ bleeders, an offset that relaxes over 4×10^{9} s and biased the stored energy in proportion to R_rad. Ramping the noise on from zero removed it, and both points then agreed within 0.5σ. The ledger itself stayed closed, because the offset was booked as energy stored at the start; the expected value computed beside the result is what exposed it.

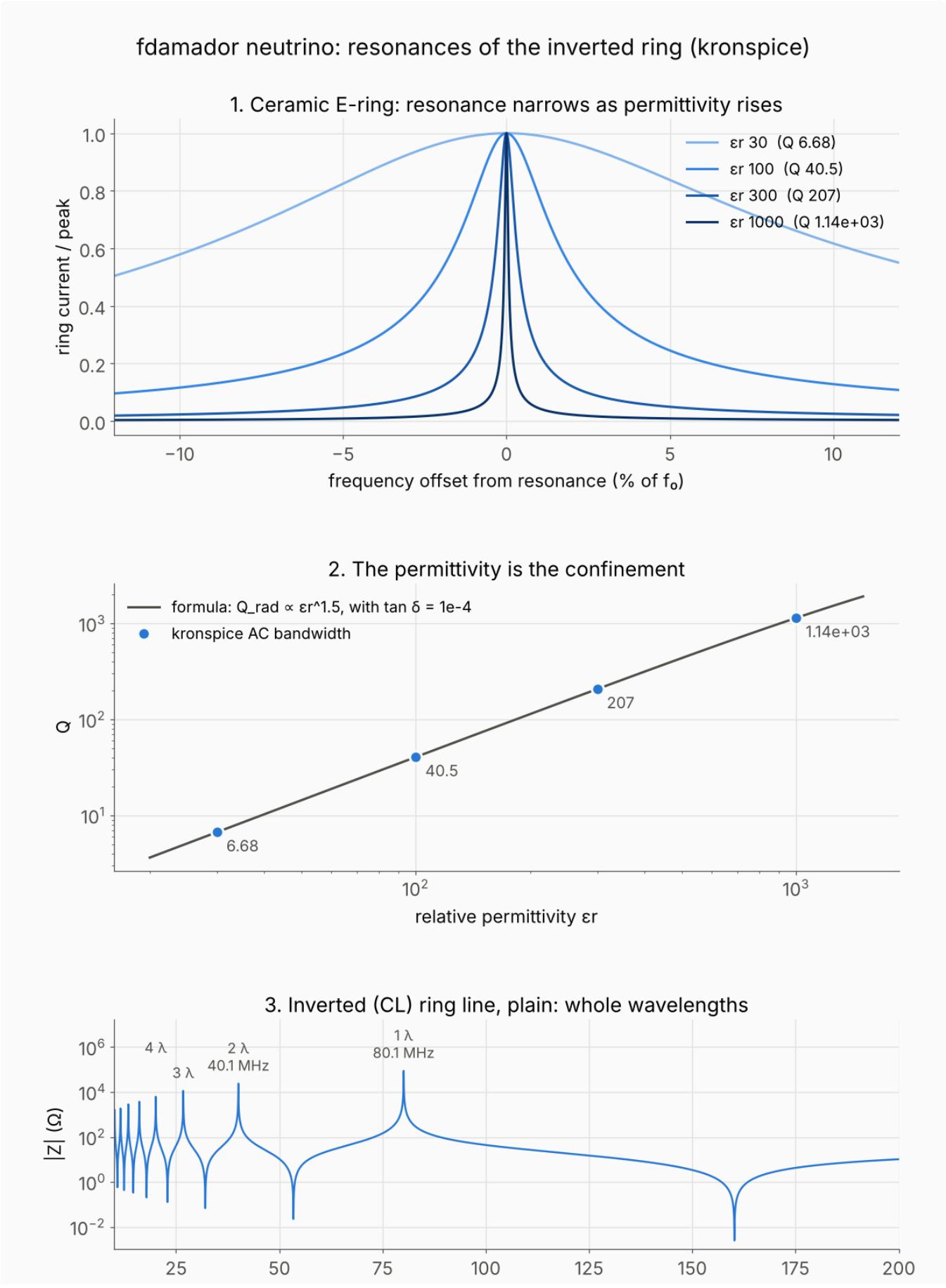
Because the ramp was added after the mismatch was seen, it was tested for whether it steers the result: on identical noise, ramps of 2Q, 5Q and 8Q agree to four digits, and an abrupt start reproduces the bias exactly. The full account, with the equations, the record of events and the check, is in Reference: the 8 % zero-point bias.

© 2026 fdamador@comcast.net

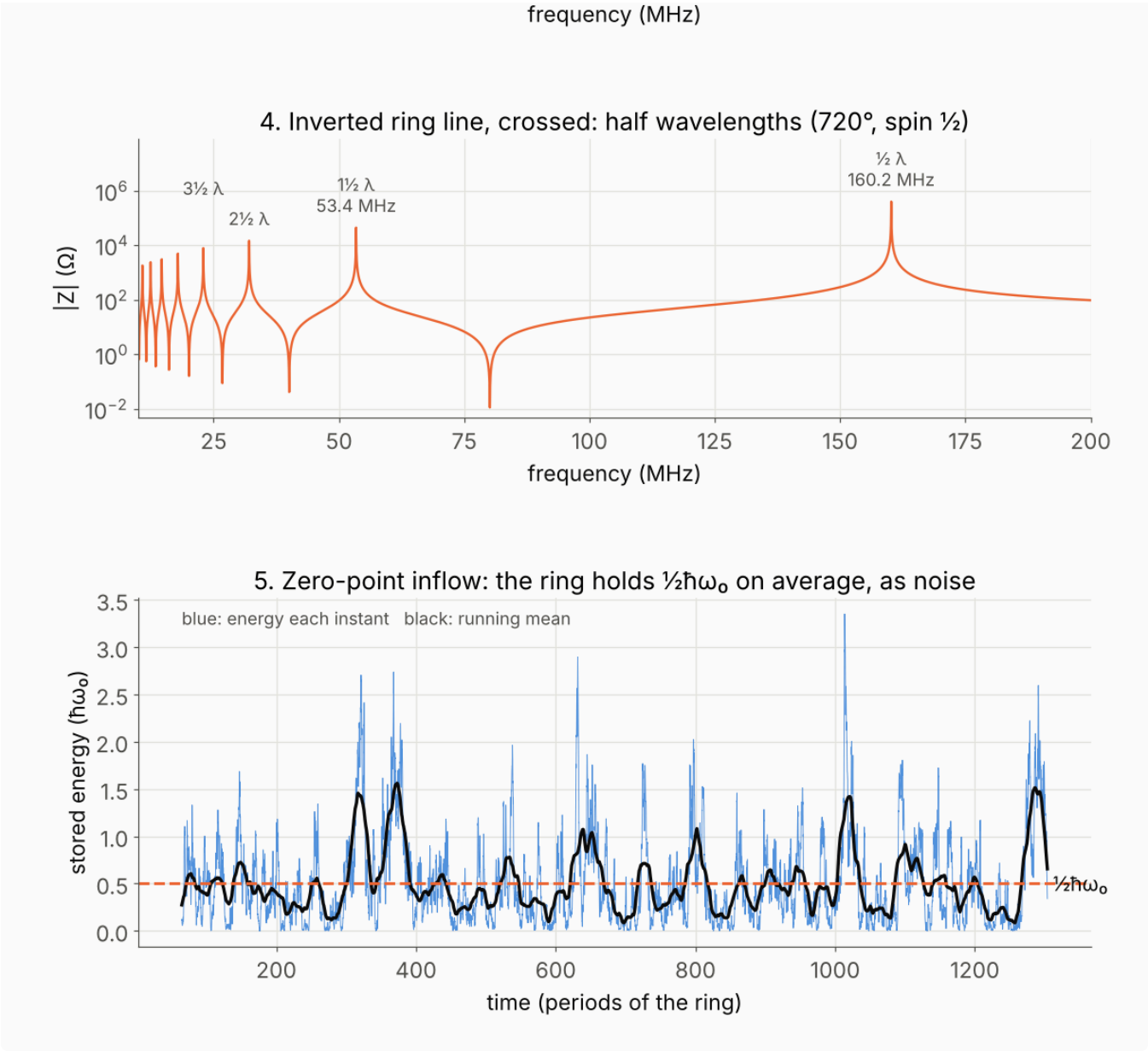
Chart and sound

The resonance chart shows five things: the E-ring's resonance narrowing as permittivity rises, Q on its εr^1.5 line, the plain and crossed ring-line spectra with each mode labelled in wavelengths, and one zero-point run with the stored energy fluctuating about ½ħω₀.

© 2026 fdamador@comcast.net



© 2026 fdamador@comcast.net



Every real frequency here is far above hearing (2.06 GHz, 20–160 MHz, 24 THz), so each sound is the kronspice waveform slowed by one factor. Every frequency ratio, decay rate and Q is kept.

Sound	Source	What you hear
1. Ceramic ring, εr 100 (Q 40.5)	kronspice ring-down	a 0.14 s tick at 440 Hz
2. Ceramic ring, εr 1000 (Q 1,140)	kronspice ring-down	a 3.4 s bell at 440 Hz

© 2026 fdamador@comcast.net

Sound	Source	What you hear
3. Ring held by the zero-point field	kronspice noise run	a 2.8 s hiss centred on 440 Hz
4. Plain inverted ring line	four highest modes, synthesized	880, 440, 294, 221 Hz: the undertones 1, ½, ⅓, ¼
5. Crossed inverted ring line	four highest modes, synthesized	880, 293, 176, 126 Hz: the odd undertones 1, ⅓, ⅕, ⅐

The harmonics file adds the ring-line modes one at a time, so each tone can be heard entering. The crossing removes every even member of the series: the spin-½ property heard rather than seen. Sounds 1–3 have no overtones, because each is a single linear resonance.

Conclusions and next steps

The inversion gets the two properties a field picture can get, no charge and no static moment, and keeps spin ½ in its circuit form; it fails on stability and has nothing to say about mass or the weak interaction.

The zero-point inflow closes the energy books but does not rescue the model. It holds every mode of every oscillator at $\frac{1}{2}\hbar\omega_0$, so it singles out no particle, and its fluctuations are as large as its mean.

What would move H28 off *bounded*:

- [ ] A confining mechanism that suppresses the ring's radiation by about 5×10^{-21}, with its own equations and a test
- [ ] An account of where the neutrino's mass comes from in the ring picture
- [ ] A coupling to the weak interaction, the only way neutrinos have ever been detected
- [ ] A prediction that differs from the standard neutrino, measurable with existing data

Repository and sources

The model lives in `examples/fdamador_neutrino` of fdamador/kronspice, on the local branch `fdamador-neutrino` (commits 4bfa244, a3ce7ba and 9e2deba; not yet pushed). All tests pass, including the zero-point inflow test. Run it with `go run`

© 2026 fdamador@comcast.net

`./examples/fdamador_neutrino`, and draw the chart and sound with `-viz` then `viz/render.py`.

Sources:

- J. G. Williamson, M. B. van der Mark, "Is the electron a photon with toroidal topology?", *Ann. Fond. Louis de Broglie* 22, 133 (1997)
- J. C. Maxwell, *Treatise on Electricity and Magnetism*, Vol. II, Art. 706 (ring self-inductance)
- C. A. Balanis, *Antenna Theory*, 3rd ed., eq. 5-24 (small-loop radiation resistance)
- D. Kajfez, P. Guillon (eds.), *Dielectric Resonators*, Artech House, 1986 (the TE01δ mode)
- H. B. Callen, T. A. Welton, *Phys. Rev.* 83, 34 (1951) (zero-point noise of a resistance)
- E. M. Purcell, *Phys. Rev.* 69, 681 (1946); R. G. Hulet, E. S. Hilfer, D. Kleppner, *PRL* 55, 2137 (1985) (boundaries change emission)
- M. Agostini et al. (Borexino), *Phys. Rev. D* 96, 091103 (2017) (neutrino magnetic moment)
- M. Aker et al. (KATRIN), *Science* 388, 180 (2025) (neutrino mass)
- Kamiokande-II and IMB, *PRL* 58, 1490 and 1494 (1987) (SN1987A neutrinos)
- H. E. Puthoff, *Phys. Rev. D* 35, 3266 (1987) (the zero-point balance of H24)

© 2026 fdamador@comcast.net